

lab-09

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Disclaimer: Generative AI was used to assist with templating and writing code in this assignment; however, this code was checked manually and edited by hand to ensure accuracy.

Source: OpenAI. (2026). *ChatGPT (GPT-5.4 Thinking)* [Large language model]. <https://chatgpt.com/>.

```
# install.packages(c("RSQLite", "DBI"))

library(RSQLite)
library(DBI)

# Initialize a temporary in memory database
con <- dbConnect(SQLite(), ":memory:")

# Download tables
doctor <- read.csv("https://raw.githubusercontent.com/dmccable/BIOSTAT620W26/main/data/hospital/doctors.csv")
visit <- read.csv("https://raw.githubusercontent.com/dmccable/BIOSTAT620W26/main/data/hospital/visits.csv")
patient <- read.csv("https://raw.githubusercontent.com/dmccable/BIOSTAT620W26/main/data/hospital/patients.csv")
payment <- read.csv("https://raw.githubusercontent.com/dmccable/BIOSTAT620W26/main/data/hospital/patient_payments.csv")

# Copy data.frames to database
dbWriteTable(con, "doctor", doctor)
dbWriteTable(con, "visit", visit)
dbWriteTable(con, "patient", patient)
dbWriteTable(con, "payment", payment)
```

```
dbListTables(con)
```

```
[1] "doctor" "patient" "payment" "visit"
```

Note: For each query, the LIMIT command was used to prevent the entire output from displaying in the knitted HTML file.

Exercise 1

```
dbGetQuery(con,
  "SELECT doctor_id, first_name, last_name
  FROM doctor
  ORDER by last_name, first_name
  LIMIT 10"
)
```

	doctor_id	first_name	last_name
1	58	CHRISTIAN	AKROYD
2	182	DEBBIE	AKROYD
3	92	KIRSTEN	AKROYD
4	118	CUBA	ALLEN
5	145	KIM	ALLEN
6	194	MERYL	ALLEN
7	76	ANGELINA	ASTAIRE
8	112	RUSSELL	BACALL
9	190	AUDREY	BAILEY
10	67	JESSICA	BAILEY

Exercise 2

```
dbGetQuery(con,
  "SELECT doctor_id, first_name, last_name
  FROM doctor
  WHERE last_name IN ('WILLIAMS', 'DAVIS')"
)
```

	doctor_id	first_name	last_name
1	4	JENNIFER	DAVIS
2	72	SEAN	WILLIAMS
3	101	SUSAN	DAVIS
4	110	SUSAN	DAVIS
5	137	MORGAN	WILLIAMS
6	172	GROUCHO	WILLIAMS

Exercise 3

```
dbGetQuery(con,
  "SELECT DISTINCT visit_id
  FROM visit
  WHERE date(visit_start_date) = '2005-07-05'
  LIMIT 10"
)
```

	visit_id
1	3470
2	3471
3	3472
4	3473
5	3474
6	3475
7	3476
8	3477

9 3478
10 3479

Exercise 4

Exercise 4.1

```
dbGetQuery(con,
  "SELECT *
  FROM payment
  WHERE amount IN (199, 799, 999)
  LIMIT 10"
)
```

	payment_id	patient_id	visit_id	amount	payment_date
1	16050	269	7	199	2007-01-24 21:40:19.996577
2	16056	270	193	199	2007-01-26 05:10:14.996577
3	16081	282	48	199	2007-01-25 04:49:12.996577
4	16103	294	595	199	2007-01-28 12:28:20.996577
5	16133	307	614	199	2007-01-28 14:01:54.996577
6	16158	316	1065	199	2007-01-31 07:23:22.996577
7	16160	318	224	999	2007-01-26 08:46:53.996577
8	16161	319	15	999	2007-01-24 23:07:48.996577
9	16180	330	967	799	2007-01-30 17:40:32.996577
10	16206	351	1137	199	2007-01-31 17:48:40.996577

Exercise 4.2

```
dbGetQuery(con,
  "SELECT *
  FROM payment
  WHERE amount > 500
  LIMIT 10"
)
```

	payment_id	patient_id	visit_id	amount	payment_date
1	16052	269	678	699	2007-01-28 21:44:14.996577
2	16058	271	1096	899	2007-01-31 11:59:15.996577
3	16060	272	405	699	2007-01-27 12:01:05.996577
4	16061	272	1041	699	2007-01-31 04:14:49.996577
5	16068	274	394	599	2007-01-27 09:54:37.996577
6	16073	276	860	1099	2007-01-30 01:13:42.996577
7	16074	277	308	699	2007-01-26 20:30:05.996577
8	16082	282	282	699	2007-01-26 17:24:52.996577

9	16086	284	1145	699	2007-01-31	18:42:11.996577
10	16087	286	81	699	2007-01-25	10:43:45.996577

Exercise 4.3

```
dbGetQuery(con,
  "SELECT *
  FROM payment
  WHERE amount > 500 AND amount < 800
  LIMIT 10"
)
```

	payment_id	patient_id	visit_id	amount		payment_date
1	16052	269	678	699	2007-01-28	21:44:14.996577
2	16060	272	405	699	2007-01-27	12:01:05.996577
3	16061	272	1041	699	2007-01-31	04:14:49.996577
4	16068	274	394	599	2007-01-27	09:54:37.996577
5	16074	277	308	699	2007-01-26	20:30:05.996577
6	16082	282	282	699	2007-01-26	17:24:52.996577
7	16086	284	1145	699	2007-01-31	18:42:11.996577
8	16087	286	81	699	2007-01-25	10:43:45.996577
9	16092	288	427	699	2007-01-27	14:38:30.996577
10	16094	288	565	599	2007-01-28	07:54:57.996577

Exercise 5

```
dbGetQuery(con,
  "SELECT payment_id, amount
  FROM payment
  INNER JOIN patient
  on payment.patient_id = patient.patient_id
  WHERE last_name = 'DAVIS'"
)
```

	payment_id	amount
1	16685	499
2	16686	299
3	16687	99

Exercise 6

Exercise 6.1

```
dbGetQuery(con,
  "SELECT COUNT(*)
  FROM visit"
)
```

```
COUNT(*)
1      16044
```

Exercise 6.2

```
dbGetQuery(con,
  "SELECT patient_id, COUNT(*)
  FROM visit
  GROUP BY patient_id
  LIMIT 10"
)
```

```
patient_id COUNT(*)
1          1      32
2          2      27
3          3      26
4          4      22
5          5      38
6          6      28
7          7      33
8          8      24
9          9      23
10         10      25
```

Exercise 6.3

```
dbGetQuery(con,
  "SELECT patient_id, COUNT(*)
  FROM visit
  GROUP BY patient_id
  ORDER BY patient_id DESC
  LIMIT 10"
)
```

```
patient_id COUNT(*)
1          599      19
2          598      22
3          597      25
4          596      28
5          595      30
6          594      27
```

7	593	26
8	592	29
9	591	27
10	590	25

Exercise 6.4

```
dbGetQuery(con,
  "SELECT patient_id, COUNT(*)
  FROM visit
  GROUP BY patient_id
  HAVING COUNT(*) >= 40
  ORDER BY patient_id DESC"
)
```

	patient_id	COUNT(*)
1	526	45
2	469	40
3	236	42
4	197	40
5	148	46
6	144	42
7	75	41

Exercise 7

```
dbGetQuery(con,
  "SELECT MAX(amount), MIN(amount), AVG(amount), SUM(amount)
  FROM payment"
)
```

	MAX(amount)	MIN(amount)	AVG(amount)	SUM(amount)
1	1199	99	416.9775	482443

Exercise 7.1

```
dbGetQuery(con,
  "SELECT MAX(patient_id), MIN(patient_id), AVG(patient_id), SUM(patient_id)
  FROM payment"
)
```

	MAX(patient_id)	MIN(patient_id)	AVG(patient_id)	SUM(patient_id)
1	599	1	292.4771	338396

Exercise 7.2

```
dbGetQuery(con,
  "SELECT MAX(patient_id), MIN(patient_id), AVG(patient_id), SUM(patient_id)
  FROM payment
  WHERE patient_id IN (
    SELECT patient_id
    FROM payment
    GROUP BY patient_id
    HAVING COUNT(*) > 5
  )"
)
```

	MAX(patient_id)	MIN(patient_id)	AVG(patient_id)	SUM(patient_id)
1	596	19	249.8977	21991

Exercise 8

Exercise 8.1

```
dbGetQuery(con,
  "SELECT doctor_id, first_name, last_name, AVG(visit_length)
  FROM (
    SELECT
      visit.doctor_id,
      first_name,
      last_name,
      julianday(visit_end_date) - julianday(visit_start_date) AS visit_length
    FROM doctor
    INNER JOIN visit
    ON doctor.doctor_id = visit.doctor_id
  )
  GROUP BY doctor_id
  LIMIT 10"
)
```

	doctor_id	first_name	last_name	AVG(visit_length)
1	1	PENELOPE	GUINNESS	4.252533
2	2	NICK	WAHLBERG	6.835764
3	3	ED	CHASE	5.737954
4	4	JENNIFER	DAVIS	6.402830
5	5	JOHNNY	LOLLOBRIGIDA	3.220862
6	6	BETTE	NICHOLSON	2.905042
7	7	GRACE	MOSTEL	7.260948
8	8	MATTHEW	JOHANSSON	3.336271

9	9	JOE	SWANK	4.341222
10	10	CHRISTIAN	GABLE	2.485816

Exercise 8.2

```
dbGetQuery(con,
  "SELECT doctor_id, first_name, last_name, AVG(visit_length)
  FROM (
    SELECT
      visit.doctor_id,
      first_name,
      last_name,
      julianday(visit_end_date) - julianday(visit_start_date) AS visit_length
    FROM doctor
    INNER JOIN visit
    ON doctor.doctor_id = visit.doctor_id
  )
  GROUP BY doctor_id
  ORDER BY AVG(visit_length) DESC
  LIMIT 5"
)
```

	doctor_id	first_name	last_name	AVG(visit_length)
1	152	BEN	HARRIS	7.652632
2	201	TOM	CRUISE	7.637725
3	50	NATALIE	HOPKINS	7.605042
4	22	ELVIS	MARX	7.500183
5	121	LIZA	BERGMAN	7.468742

```
dbGetQuery(con,
  "SELECT doctor_id, first_name, last_name, AVG(visit_length)
  FROM (
    SELECT
      visit.doctor_id,
      first_name,
      last_name,
      julianday(visit_end_date) - julianday(visit_start_date) AS visit_length
    FROM doctor
    INNER JOIN visit
    ON doctor.doctor_id = visit.doctor_id
  )
  GROUP BY doctor_id
  ORDER BY AVG(visit_length) ASC
  LIMIT 5"
)
```

	doctor_id	first_name	last_name	AVG(visit_length)
1	93	ELLEN	PRESLEY	2.213376

2	182	DEBBIE	AKROYD	2.368833
3	10	CHRISTIAN	GABLE	2.485816
4	14	VIVIEN	BERGEN	2.514360
5	187	RENEE	BALL	2.521940

```
# clean up  
dbDisconnect(con)
```